

Algebra 1
Unit 5
Lesson 11 Practice

Name Answer Key
Date _____
Period _____

Tyler owns a food truck that sells chicken wings and hamburger sliders. Each order of chicken wings sells for \$7.25 while each order of hamburger sliders sells for \$3. On Friday, Tyler made a total of \$595 in revenue from all sliders and wings sales. There were twice as many chicken wing orders sold than there were sliders sold.

1. Define the variables to write the system of equations.

Let c = chicken wing orders sold

Let s = sliders sold

2. Write a system of equations to determine the number of chicken wing orders sold and the number of hamburger sliders sold.

$$\begin{cases} 7.25c + 3s = 595 \\ c = 2s \end{cases}$$

3. Solve the system using the method of your choice.

$$\begin{array}{ll} 7.25(2s) + 3s = 595 & c = 2(34) \\ 14.5s + 3s = 595 & c = 68 \\ 17.5s = 595 & \\ s = 34 & \end{array}$$

4. The number of chicken wing orders on Friday was 68. The number of hamburger slider orders was 34.

Logan has \$0.87 worth of pennies and nickels. He has a total of 31 coins altogether. Determine the number of pennies and nickels Logan has.

5. Write a system of equations that could be used to determine the number of pennies and nickels Logan has.

$$\begin{cases} 0.01p + 0.05n = 0.87 \\ p + n = 31 \end{cases}$$

6. Solve the system of equations.

$$\begin{array}{ll} p = 31 - n & \\ 0.01(31 - n) + 0.05n = 0.87 & p + n = 31 \\ 0.31 - 0.01n + 0.05n = 0.87 & p + 14 = 31 \\ 0.04n = 0.56 & p = 17 \\ n = 14 & \\ 17 \text{ pennies} & \\ 14 \text{ nickels} & \end{array}$$

Axel wanted to make salsa, so he went to the grocery store and bought 7 tomatoes, and 4 onions and spent \$17.75. He didn't buy enough so he went to the same store and bought 5 more tomatoes and 2 more onions spending \$10.75.

7. Write a system of equations that could be used to find the price of tomatoes and onions.

$$\begin{cases} 7t + 4x = 17.75 \\ 5t + 2x = 10.75 \end{cases}$$

8. Solve the system of equations.

$$\begin{array}{rcl} -2(5t + 2x = 10.75) & \rightarrow & -10t - 4x = -21.5 \\ 7t + 4x = 17.75 & & 7(1.25) + 4x = 17.75 \\ \hline -10t - 4x = -21.5 & & 8.75 + 4x = 17.75 \\ -3t = -3.75 & & 4x = 9 \\ t = 1.25 & & x = 2.25 \end{array}$$

Tomatoes: \$1.25

Onions: \$2.25

Pressley is working two part-time jobs. He makes \$18 per hour lifeguarding and \$12 per hour as a cashier at the local grocery store. During spring break from school, he worked twice as many hours lifeguarding as he did at the local grocery store and earned \$144. How many hours did he work at each job?

9. Write a system of equations that could be used to determine the number of hours Pressley worked.

$$\begin{cases} 18x + 12c = 144 \\ x = 2c \end{cases}$$

10. Solve the system using the method of your choice.

$$\begin{array}{rcl} 18(2c) + 12c = 144 & & x = 2(3) \\ 36c + 12c = 144 & & x = 6 \\ 48c = 144 & & \\ c = 3 & & \end{array}$$

6 hours lifeguarding

3 hours as a cashier